Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

MIX MECHS

Project / Solution Name

SkillFolio

Name (Leader)

Jaiydaan Raj

Email & Phone (Leader)

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University Affiliation(s)

University of Southampton Malaysia

Live Prototype URL

https://skillfolio.replit.app/

Code Repository URL (public / access granted to judges)

https://github.com/SDC-HACKATHON-MIX-MECHS/skillfolio (public)

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The user opens SkillFolio, a Streamlit web app hosted on Replit, on a phone or laptop. On the Profile page they paste a made-up resume or pick skills. The app finds skill names by keyword matching. When the two RunPod keys are set, it also sends the text once to our RunPod Serverless GPU endpoint. The public demo runs without the keys. There, the open model BAAI/bge-small-en-v1.5 compares each sentence with a description of each of our 73 skills and returns the closest skill ids. The app only accepts ids from its own list, and falls back to keyword matching if the endpoint is slow or down.

The Python engine then reads a dated data snapshot built by our pipeline from Hacker News "Who is hiring" posts (via the HN Algolia API), the Anthropic Economic Index and O*NET. It forecasts demand for each skill under two AI scenarios, scores the user's skills, ranks 20 roles by fit, picks free courses from our checked list, and links roles to UN SDGs. The user's course progress stays in the browser session and updates their skill levels. All data files are versioned in the public GitHub repository and tested automatically.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Language for the app, engine, pipeline and RunPod worker	PSF License
Streamlit 1.66	Web app and pages	Apache-2.0
pandas 3.0	Data tables in the engine and pipeline	BSD-3-Clause
NumPy 2.5	Forecast and portfolio maths; similarity scores	BSD-3-Clause
Plotly 7.1	Charts	MIT
Requests 2.34	Calls to the HN Algolia API and the RunPod endpoint	Apache-2.0
pypdf 6.21	Reading text from uploaded PDF resumes	BSD-3-Clause
sentence-transformers 6.2	Running the embedding model on RunPod	Apache-2.0
PyTorch 2.x	Model runtime on the RunPod GPU	BSD-3-Clause
RunPod Python SDK 1.x	Serverless worker on RunPod	MIT
pytest 9.1	Automated tests (over 900)	MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
HN Algolia Search API (Algolia / Hacker News)	Downloads monthly "Who is hiring" job posts for skill demand trends	Free, public
RunPod Serverless API (RunPod)	Calls our own embedding-model endpoint for resume-to-skills matching	Sponsor-provided credits

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Replit Inc.)	Shared development workspace; public hosting of the live app; Secrets for the RunPod keys	Replit Core (hackathon-provided); Deployments
RunPod Serverless (RunPod)	Hosts the embedding model; built from our GitHub repo's Dockerfile	Sponsor credits; 16 GB GPU, 0 always-on / 1 max worker; global data centres
GitHub (GitHub Inc.)	Source code, data snapshots, version control	Free, public repository

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
BAAI/bge-small-en-v1.5	Beijing Academy of Artificial Intelligence, via Hugging Face	Pre-trained (not fine-tuned); used as-is to embed resume sentences and skill descriptions	MIT
Skill demand forecast (log-linear trend with shrinkage and scenario shift)	Our team (engine/forecast.py)	Self-built statistical model; no training on personal data	Apache-2.0 (our repo)

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
UN SDG targets 4.4 and 8.5	https://unstats.un.org/sdgs/metadata/	Public	UN public data	N
Skills, roles and course catalogue (73 skills, 20 roles, 52 free courses)	Built by our team from the sources above and course providers' public pages	Self-collected	Apache-2.0 (our repo)	N
Hacker News "Ask HN: Who is hiring?" posts, Oct 2024 – Oct 2026 (7,961 posts)	https://hn.algolia.com/api	Public	Public API	N (only skill counts are stored)
Anthropic Economic Index, job exposure	https://www.anthropic.com/research/labor-market-impacts	Public	CC-BY	N
O*NET 31.0 Technology Skills	https://www.onetcenter.org/database.html	Public	CC BY 4.0	N
Five sample profiles (Aina, Hafiz, Mei, Priya, Ravi)	Written by our team	Synthetic	Apache-2.0 (our repo)	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code (Anthropic; Claude Opus and Sonnet models)	Writing and testing code, data pipeline, UI, documentation and draft text. No other generative-AI tool was used	Major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every change went through a GitHub pull request that a team member read. We wrote automated tests first (over 900 now) covering data formats, every engine formula against separately worked numbers, and app flows; AI-written code had to pass them. A second Claude Code session that could not edit anything re-ran the tests and checked the app before we called work done. A team member checked each number, course and source against its original page. The team is responsible for all submitted code and text.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	Cloud credits (USD 25)	Serverless GPU endpoint running the embedding model for resume-to-skills matching
Replit	Replit Core plan	Development workspace and public deployment of the live prototype

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to sharing the prototype: it hosts the live app judges use. RunPod runs the AI model for the resume step (finding skills a user describes without naming them) when its keys are set; the public demo runs without it. The rest of the app (forecasts, matching, plans, SDG links) does not need it, and the resume step falls back to keyword matching. The model could move to another GPU host, but RunPod's per-second billing keeps idle cost near zero.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

SkillFolio has no accounts and no user database. We ask users not to enter a real resume, and the demo uses made-up profiles. Resume text is read in memory, used once to find skills, then thrown away: it is not written to disk, logged, or kept in the session. In the public demo no resume text leaves the app's server. When the RunPod keys are set and the user leaves "Also use our AI model" ticked, the text is sent once over HTTPS to our own RunPod endpoint, which returns skill ids and keeps nothing. The saved profile holds only skill ids, levels, hours, budget, risk comfort and chosen SDGs. It lives in the browser session and disappears when the session ends. The app never asks for age, gender or other protected details. API keys are kept in Replit Secrets and a git-ignored local file.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The AI model can't invent skills: it only ranks ids from our list, and the app rejects anything else. Users confirm or remove every suggestion. It sometimes suggests too much, for example all three cloud providers for "a cloud server". Our 20 roles and 73 skills are tech and data, so other industries aren't covered. Hacker News posts are mostly US and remote tech jobs, not the Malaysian market, and skills with too few posts get no trend. Users tick their own course progress, and a very short course can raise the fit score a lot. Portfolios last one session. Forecasts are estimates under two scenarios. A cold RunPod worker can take up to a minute to start; after 40 seconds the app uses keyword matching.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date]

25/10/26